

Generative models of rotor speed and rotor noise

The fitted trajectory model, the fitted noise model, and what they transfer

Dmitrii Mukhutdinov

2026-09-22

$$w(t) = \mu + \delta_{\text{flight}} + M R(\theta) v(t) + e(t)$$

$$v_i(t) = \text{OU}(\tau_{\text{slow},i}, \sigma_{\text{slow},i}) + \text{CAR2}(f_{0,i}, \zeta_i, \sigma_{\text{osc},i})$$

$$e_j(t) = \text{OU}(\tau_e, \sigma_e), \quad \text{independent across rotors}$$

$$\delta_{\text{flight}} = \mathbf{1}c + r, \quad c \sim N(0, s_c^2), \quad r \sim N(0, \text{diag}(s_r^2))$$

$$\text{CAR2} : \ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = \xi, \quad \omega_0 = 2\pi f_0$$

M = mixer, columns [common, roll, pitch, yaw]; $R(\theta)$ rotates the (common, yaw) plane.

$$S(\omega) = \frac{2}{f_s} H(e^{i\omega} I - \Phi)^{-1} Q (e^{-i\omega} I - \Phi^\top)^{-1} H^\top, \quad \Phi = e^{A\Delta t}, \quad Q = P - \Phi P \Phi^\top.$$

Trajectory model — parameters

symbol	meaning	shape	range / prior
μ	per-rotor hover level (rev/s)	4	estimated from airborne samples
θ	rotation of the (collective, yaw) plane (rad)	1	$(-\pi/4, \pi/4]$ canonical
τ_{slow}	slow OU time constant per mode (s)	4	$(\tau_c, 10]$, LN(2.0 s, 1.5)
σ_{slow}	slow OU stationary sd per mode (rev/s)	4	LN(1.0, 3.0)
f_0	oscillator centre frequency per mode (Hz)	4	(0.05, 20)
ζ	oscillator damping ratio per mode	4	(0.2, 3.0)
σ_{osc}	oscillator stationary sd per mode (rev/s)	4	LN(1.0, 3.0)
τ_e	measurement OU time constant (s)	1	$(10^{-3}, 1)$, LN(0.05 s, 2.0)
σ_e	measurement OU sd, per rotor (rev/s)	1	LN(0.1, 3.0)
s_c	sd of the per-flight level common to the rotors (rev/s)	1	moment estimate
s_r	sd of the per-flight per-rotor level (rev/s)	4	moment estimate

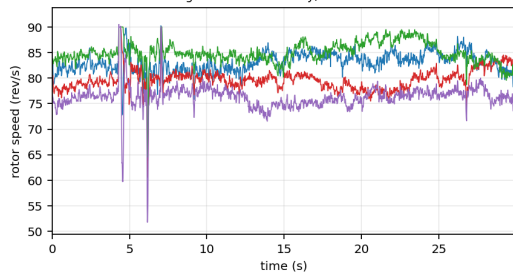
Trajectory data

rig	vehicle	flights	airborne (s)	native rate	speed source
michaels	DJI Matrice 100	4	395	29.41 Hz	DatCon ESC feedback
dregon	MikroKopter quad	10	429	~1 kHz grid	motor.measured
neurobem_quad	custom quad	247	2479	400 Hz	Betaflight ESC feedback
pitcn_quad	micro quad, 0.25 kg	68	3345	100 Hz	ESC telemetry
nanobench_cf21b	Crazyflie 2.1 Brushless	15	668	100 Hz	bidirectional DSHOT eRPM
vid_m100	DJI M100 derivative	4	397	~1 kHz CAN	M3508/C620 CAN feedback
blackbird_quad	custom quad	1	199	187 Hz	optical motor encoders
total		349	7912		

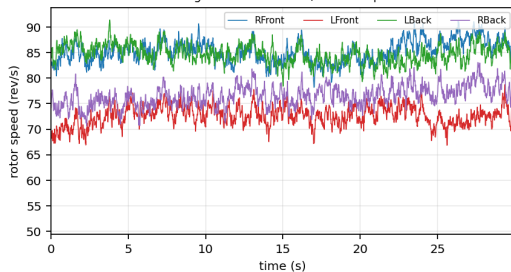
Hover level 78–278 rev/s; per-rotor variance spans a factor of 600. Common analysis grid 100 Hz.

Real vs sampled trajectories

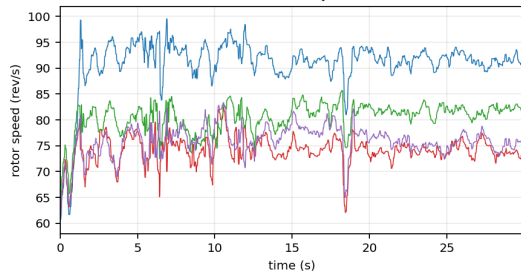
dragon — real telemetry, 30 s airborne



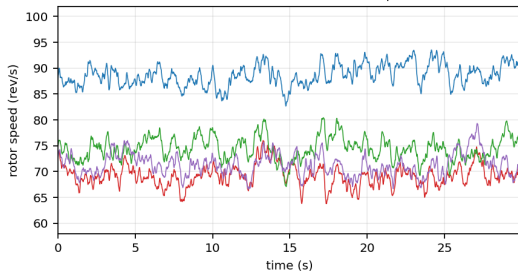
dragon — fitted model, 30 s sample



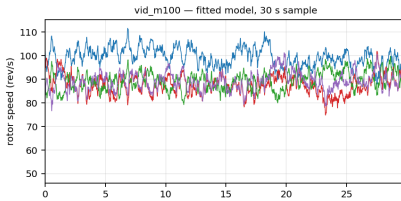
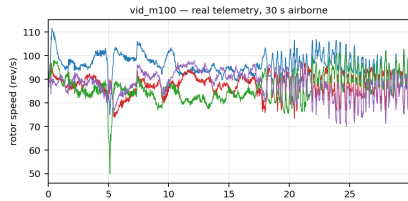
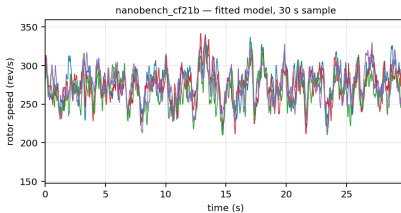
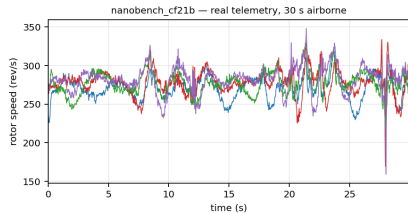
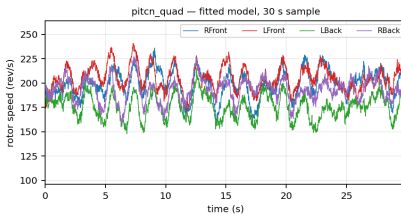
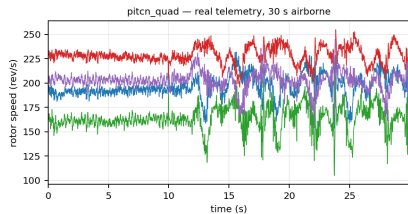
michaels — real telemetry, 30 s airborne



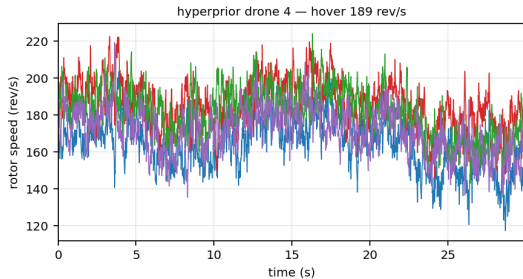
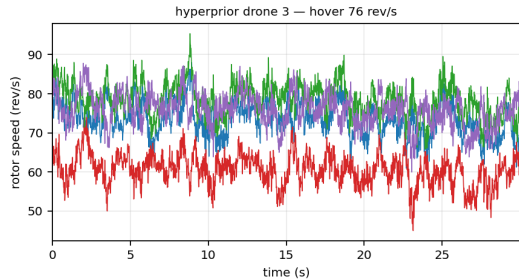
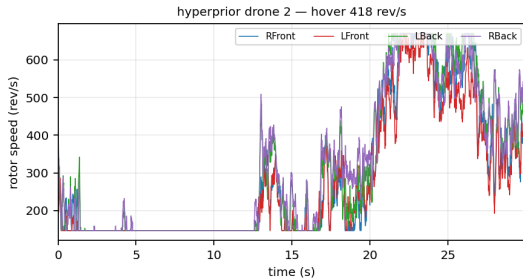
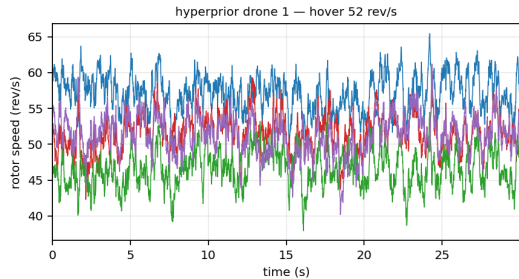
michaels — fitted model, 30 s sample



Real vs sampled trajectories — other rigs



Trajectories from the rig hyperprior



$$x_m(t) = \sum_r \sum_k A_{mrk}(t) \cos \phi_{mrk}(t) + n_m(t), \quad \phi_{mrk} = 2\pi k \int_0^t f_r + k\theta_r(t) + \psi_{rk}(t) + \alpha_{mrk}$$

$$d\nu_r = -\lambda\nu_r dt + \sqrt{2\lambda}\sigma_\nu dW_r, \quad d\theta_r = \nu_r dt, \quad V_\theta(\tau) = 2\sigma_\nu^2 \left[\frac{|\tau|}{\lambda} - \frac{1-e^{-\lambda|\tau|}}{\lambda^2} \right]$$

$$\text{Var}[\Delta\psi_{rk}] = 4\pi\gamma_{rk}|\tau| \Rightarrow R_{rk}(\tau) = \exp\left[-\frac{k^2}{2}V_\theta(\tau) - 2\pi\gamma_{rk}|\tau|\right]$$

$$A_{mrk}(t) = \sqrt{2P_{rk}} \left(\frac{f_r(t)}{f_{\text{ref}}} \right)^{a/2} 10^{(g_{mr}-\bar{g})/20} 10^{(G_m-\bar{G})/20}, \quad P_{rk} = 10^{p_{rk}/10}$$

$$S_m(f,t) = 10^{h_m/10} \left[\frac{1}{R} \sum_r \left(\frac{f_r(t)}{f_{\text{ref}}} \right)^b + s \right] 10^{(\mu_F + \sum_{j=1}^{14} B_j(f) \sigma_B(Lz)_j + t_F \log_2(f/f_t))/10}$$

Regimes (Michael's): standby and cruise fits composed on the carrier; the blend weight follows $f_r(t)$.

Noise model — parameters

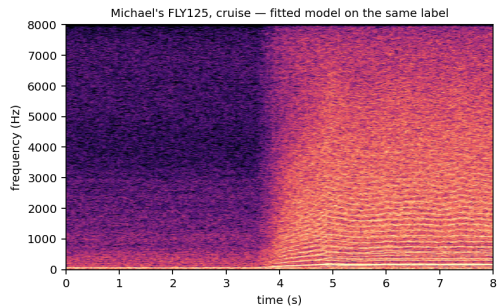
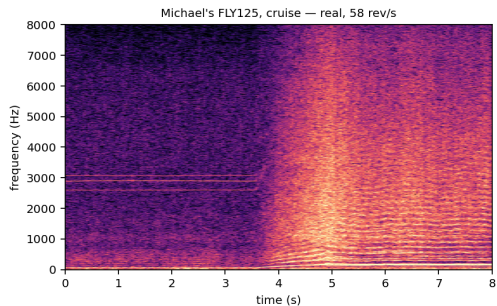
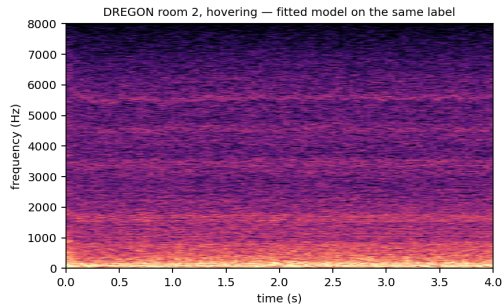
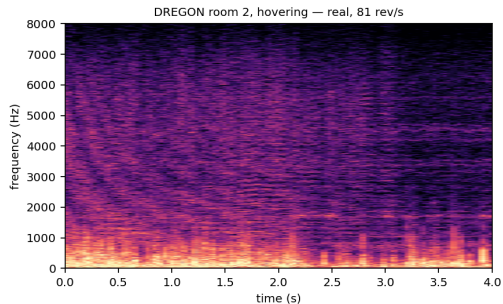
symbol	meaning	shape	prior
σ_v	shaft OU speed-error sd (rad/s)	per rotor	bench LN(0.3, 0.7); flight LN(0.3, 0.5)
λ	shaft OU rate (s^{-1})	per rotor	LN(2, 1.0) on the bench; pinned in flight
γ_{rk}	Lorentzian half-width of line (r, k) (Hz)	rotor $\times$ order	LN(0.01 k Hz, 1.0), $\gamma \geq 0$
p_{rk}	line power of order k (dB) — the comb profile	rotor $\times$ order	N(measured, 10) if SNR ≥ 2 dB, else N($\mu_F - 15$, 8)
c	level of a transplanted comb (dB)	1	N(0, 20)
a	comb speed exponent	1	N(2, 1); pinned at 2 on a short span
b	floor speed exponent	1	LN(2, 0.5); same span pin
s	static fraction of the floor	1	LN(2.5×10^{-3} , 1.0)
μ_F	floor level (dB)	1	N(measured band median, 10)
z	GP control points of the floor shape	14	N(0, 1) each
t_F	floor tilt (dB/octave)	1	N(0, 5)
g_{mr}, h_m, G_m	per-mic line / floor / overall gain (dB)	mic $\times$ rotor, mic, mic	N(0, 6)
f_r	carrier (rev/s)	per rotor	given: window-refined or telemetry label

Noise data

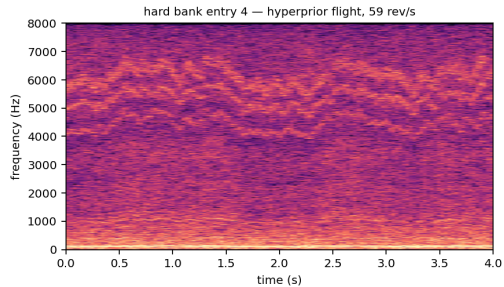
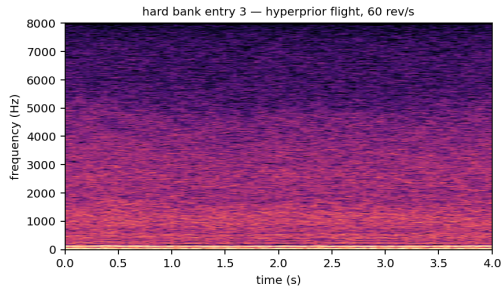
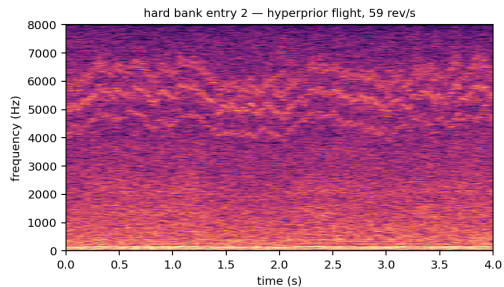
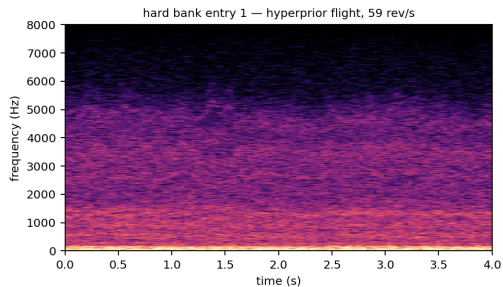
recording set	class	rotors / mics	duration	role in the fit
DREGON single-motor bench, 50–90 %	bench	1 / 8	12–34 s per window	identifies σ_ν , λ , γ_{rk}
DREGON motor_allMotors_70 bench	bench	4 / 8	~30 s	four-rotor control
DREGON room 2: free flight, hover, rectangle, spinning	flight	4 / 8	4 s frames, pooled	floor and flight profile (R5 fit)
Michael's FLY125 cruise	flight	4 / 1	4 s frames, pooled	cruise comb and floor (R3 fit)
Michael's FLY125 standby	flight	4 / 1	4 s frames, pooled	standby regime of the same rig

Flight supports: periodic Hann, $N = 2048$, hop 512 at 16 kHz. Fit band 30 Hz – 7900 Hz.

Real vs sampled noise, on the same label



Hard-bank noise (noise-fit path $\times$ trajectory hyperprior) used for hard training



Results — rev/s MAE on the real validation split (regime-only)

trunk	arm	status	real_overall	zero	standby	ramp	cruise
SCv2	real_r4_scv2	final	2.99	2.44	3.20	8.17	2.95
	(real ref)						
SCv2	nv2_easy_scv2	final	7.94	10.64	11.05	11.12	6.87
	(synthetic only)						
SCv2	nv2_hard_scv2	final	7.06	1.25	6.16	15.81	7.76
	(synthetic only)						
SCv2	nv2_mixed_scv2	final	2.53	—	—	—	—
	(mixed)						
SCv2	nv2_easy_ft	final	2.73	2.23	2.36	8.16	2.60
	(easy → real)						
SCv2	nv2_hard_ft	running, r32	2.23	—	—	—	—
	(hard → real)						
HPPNet L2	hppnet_12_r2	final	2.27	—	—	—	—
	(real ref)						
HPPNet L2	nv2_easy_hppnet_12	final	6.46	3.58	5.04	26.50	6.14
	(synthetic only)						
HPPNet L2	nv2_hard_hppnet_12	running, r93	6.80	—	—	—	—
	(synthetic only)						

Conclusions

- ▶ We have a probabilistic generative model of drone ego-noise — not deep, built on Gaussian processes — that is fitted to real data points, both for the drone **trajectories** and for the audio those trajectories produce.
- ▶ The model is fitted on a small amount of real data from several drones, and the resulting posterior over model parameters lets us sample **unseen** drones.
- ▶ Models trained only on synthetic data obtained this way still produce meaningful predictions on real data.
- ▶ Models trained on **both** real and synthetic data reach **better overall performance** than models trained on real data only.